Effects of an overnight intravenous lipid infusion on intramyocellular lipid content and insulin sensitivity in African-American versus Caucasian adolescents

Objective To explain the predisposition for insulin resistance among African American (AA) adolescents, this study aimed to: 1) examine changes in intramyocellular lipid content (IMCL), and insulin sensitivity with intralipid (IL) infusion; and 2) determine whether the increase in IMCL is comparable between AA and Caucasian adolescents. Materials and Methods Thirteen AA and 15 Caucasian normal-weight adolescents (BMI <85th) underwent a 3-h hyperinsulinemic-euglycemic clamp, on two occasions in random order, after an overnight 12-hr infusion of: 1) 20% IL and 2) normal saline (NS). IMCL was quantified by 1H-magnetic resonance spectroscopy in tibialis anterior muscle before and after IL infusion. Results During IL infusion, plasma TG, glycerol, FFA and fat oxidation increased significantly, with no race differences. Hepatic insulin sensitivity decreased with IL infusion with no difference between the groups. IL infusion was associated with a significant increase in IMCL, which was comparable between AA (Δ 105%; NS: 1.9 ± 0.8 vs. IL: 3.9 ± 1.6 mmol/kg wet weight) and Caucasian (Δ 86%; NS: 2.8 ± 2.1 vs. IL: 5.2 ± 2.4 mmol/kg wet weight), with similar reductions ($P<0.01$) in insulin sensitivity between the groups (Δ –44%: NS: 9.1 ± 3.3 vs. IL: 5.1 ± 1.8 mg/kg/min per µU/ml in AA) and (Δ–39%: NS: 12.9 ± 6.0 vs. IL: 7.9 ± 3.8 mg/kg/min per µU/ml in Caucasian) adolescents. Conclusions In healthy adolescents, an acute elevation in plasma FFA with IL infusion is accompanied by significant increases in IMCL and reductions in insulin sensitivity with no race differential. Our findings suggest that AA normal-weight adolescents are not more susceptible than Caucasians to FFA-induced IMCL accumulation and insulin resistance.